

Card 1

Q1. Discrete random variables. Examples

Q2. p-value, t-Test. Examples

Q3. A certain polymer is used for evacuation systems for aircraft. It is important that the polymer be resistant to the aging process. Twenty specimens of the polymer were used in an experiment. Ten were assigned randomly to be exposed to an accelerated batch aging process that involved exposure to high temperatures for 10 days. Measurements of tensile strength of the specimens were made, and the following data were recorded on tensile strength in psi:

No aging:	227	222	218	217	225
	218	216	229	228	221
Aging:	219	214	215	211	209
	218	203	204	201	205

(a) Calculate the sample mean tensile strength of the two samples.

(b) Calculate the median for both.

Q4. A witness to a hit-and-run accident told the police that the license number contained the letters RLH followed by 3 digits, the first of which was a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

Q5. A UCLA researcher claims that the life span of mice can be extended by as much as 25% when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diet is enriched to normal levels by vitamins and protein.

Assuming that it is known from previous studies that $\sigma = 5.8$ months, how many mice should be included in our sample if we wish to be 99% confident that the mean life span of the sample will be within 2 months of the population mean for all mice subjected to this reduced diet?

Q6. The heights of 1000 students are approximately normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Suppose 200 random samples of size 25 are drawn from this population and the means recorded to the nearest tenth of a centimeter. Determine

(a) the mean and standard deviation of the sampling distribution of $\bar{X}$;

(b) the number of sample means that fall between 172.5 and 175.8 centimeters inclusive;

(c) the number of sample means falling below 172.0 centimeters.

Card 2

Q1. Sample space. Examples

Q2. The Central Limit Theorem. Examples

Q3. The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint.

3.4	2.5	4.8	2.9	3.6
2.8	3.3	5.6	3.7	2.8
4.4	4.0	5.2	3.0	4.8

Assume that the measurements are a simple random sample.

- (a) What is the sample size for the above sample?
- (b) Calculate the sample mean for these data.
- (c) Calculate the sample median.
- (d) Compute the 20% trimmed mean for the above dataset.

Q4. Find the errors in each of the following statements:

(a) The probabilities that an automobile salesperson will sell 0, 1, 2, or 3 cars on any given day in February are, respectively, 0.19, 0.38, 0.29, and 0.15.

(b) The probabilities that a printer will make 0, 1, 2, 3, or 4 or more mistakes in setting a document are, respectively, 0.19, 0.34, -0.25, 0.43, and 0.29.

(c) On a single draw from a deck of playing cards, the probability of selecting a heart is $1/4$, the probability of selecting a black card is $1/2$, and the probability of selecting both a heart and a black card is $1/8$.

Q5. The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X;
- (b) using the probability density function of X.

Q6. A study was conducted at Virginia Tech to determine if certain static arm-strength measures have an influence on the “dynamic lift” characteristics of an individual. Twenty-five individuals were subjected to strength tests and then were asked to perform a weight lifting test in which weight was dynamically lifted over head. The data are given here.

Individual	Arm Strength, x	Dynamic Lift, y
1	17.3	71.7
2	19.3	48.3
3	19.5	88.3
4	19.7	75.0
5	22.9	91.7
6	23.1	100.0
7	26.4	73.3
8	26.8	65.0
9	27.6	75.0
10	28.1	88.3
11	28.2	68.3
12	28.7	96.7
13	29.0	76.7
14	29.6	78.3
15	29.9	60.0
16	29.9	71.7
17	30.3	85.0
18	31.3	85.0
19	36.0	88.3
20	39.5	100.0
21	40.4	100.0
22	44.3	100.0
23	44.6	91.7
24	50.4	100.0
25	55.9	71.7

- (a) Estimate β_0 and β_1 for the linear regression curve $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (b) Find a point estimate of $\mu_{Y|30}$.

Card 3

Q1. Interval Estimates. Examples

Q2. The sample mean. Examples

Q3. Suppose that in a senior college class of 500 students it is found that 210 smoke, 258 drink alcoholic beverages, 216 eat between meals, 122 smoke and drink alcoholic beverages, 83 eat between meals and drink alcoholic beverages, 97 smoke and eat between meals, and 52 engage in all three of these bad health practices. If a member of this senior class is selected at random, find the probability that the student

- (a) smokes but does not drink alcoholic beverages;
- (b) eats between meals and drinks alcoholic beverages but does not smoke;
- (c) neither smokes nor eats between meals.

Q4. Twelve people are given two identical speakers, which they are asked to listen to for differences, if any. Suppose that these people answer simply by guessing. Find the probability that three people claim to have heard a difference between the two speakers.

Q5. The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random variable X having a continuous uniform distribution with $A = 7$ and $B = 10$. Find the probability that on a given day the amount of coffee dispensed by this machine will be

- (a) at most 8.8 liters;
- (b) more than 7.4 liters but less than 9.5 liters;
- (c) at least 8.5 liters.

Q6. The amounts of a chemical compound y that dissolved in 100 grams of water at various temperatures x were recorded as follows:

x ($^{\circ}\text{C}$)	y (grams)		
0	8	6	8
15	12	10	14
30	25	21	24
45	31	33	28
60	44	39	42
75	48	51	44

- (a) Find the equation of the regression line.
- (b) Graph the line on a scatter diagram.
- (c) Estimate the amount of chemical that will dissolve in 100 grams of water at 50°C .

Card 4

Q1. Inferences for regression coefficients. Examples

Q2. Exponential Distribution. Examples

Q3. The lengths of time, in minutes, that 10 patients waited in a doctor's office before receiving treatment were recorded as follows: 5, 11, 9, 5, 10, 15, 6, 10, 5, and 10. Treating the data as a random sample, find

- (a) the mean;
- (b) the median;
- (c) the mode.

Q4. A box contains 500 envelopes, of which 75 contain \$100 in cash, 150 contain \$25, and 275 contain \$10. An envelope may be purchased for \$25. What is the sample space for the different amounts of money? Assign probabilities to the sample points and then find the probability that the first envelope purchased contains less than \$100.

Q5. The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

Q6. The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters.

- (a) Construct a 98% confidence interval for the mean height of all college students.
- (b) What can we assert with 98% confidence about the possible size of our error if we estimate the mean height of all college students to be 174.5 centimeters?

Card 5

Q1. Properties of probability.

Q2. The sample variance. Examples

Q3. Twenty adult males between the ages of 30 and 40 participated in a study to evaluate the effect of a specific health regimen involving diet and exercise on the blood cholesterol. Ten were randomly selected to be a control group, and ten others were assigned to take part in the regimen as the treatment group for a period of 6 months. The following data show the reduction in cholesterol experienced for the time period for the 20 subjects: Compute the mean, median, and 10% trimmed mean for both groups.

Control group:	7	3	-4	14	2
	5	22	-7	9	5
Treatment group:	-6	5	9	4	4
	12	37	5	3	3

Q4. The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X;
- (b) using the probability density function of X.

Q5. A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
- (b) what is the probability that a cup contains between 191 and 209 milliliters?
- (c) how many cups will probably overflow if 230- milliliter cups are used for the next 1000drinks?
- (d) below what value do we get the smallest 25% of the drinks?

Q6. A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:

Temperature, x	Converted Sugar, y
1.0	8.1
1.1	7.8
1.2	8.5
1.3	9.8
1.4	9.5
1.5	8.9
1.6	8.6
1.7	10.2
1.8	9.3
1.9	9.2
2.0	10.5

- (a) Estimate the linear regression line.
- (b) Estimate the mean amount of converted sugar produced when the coded temperature is 1.75.

Card 6

Q1. Definition of combinations and calculation formula.

Q2. Tests on two means. Examples

Q3. The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate

(a) in both cities?

(b) in neither city?

Q4. According to Chemical Engineering Progress approximately 30% of all pipework failures in chemical plants are caused by operator error.

(a) What is the probability that out of the next 20 pipework failures at least 10 are due to operator error?

(b) What is the probability that no more than 4 out of 20 such failures are due to operator error?

Q5. A random sample of 100 automobile owners in the state of Virginia shows that an automobile is driven on average 23,500 kilometers per year with a standard deviation of 3900 kilometers. Assume the distribution of measurements to be approximately normal.

(a) Construct a 99% confidence interval for the average number of kilometers an automobile is driven annually in Virginia.

(b) What can we assert with 99% confidence about the possible size of our error if we estimate the average number of kilometers driven by car owners in Virginia to be 23,500 kilometers per year?

Q6. The proportion of adults living in a small town who are college graduates is estimated to be $p = 0.6$. To test this hypothesis, a random sample of 15 adults is selected. If the number of college graduates in the sample is anywhere from 6 to 12, we shall not reject the null hypothesis that $p = 0.6$; otherwise, we shall conclude that $p \neq 0.6$.

(a) Evaluate α assuming that $p = 0.6$.

(b) Evaluate β for the alternatives $p = 0.5$ and $p = 0.7$.

Card 7

Q1. Theorem of multiplication of probabilities.

Q2. Properties of least squares estimators. Examples

Q3. An automobile manufacturer is concerned about a possible recall of its best-selling four-door sedan. If there were a recall, there is a probability of 0.25 of a defect in the brake system, 0.18 of a defect in the transmission, 0.17 of a defect in the fuel system, and 0.40 of a defect in some other area.

(a) What is the probability that the defect is the brakes or the fueling system if the probability of defects in both systems simultaneously is 0.15?

(b) What is the probability that there are no defects in either the brakes or the fueling system?

Q4. The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

(a) less than 120 hours;

(b) between 50 and 100 hours.

Q5. A common practice of airline companies is to sell more tickets for a particular flight than there are seats on the plane, because customers who buy tickets do not always show up for the flight. Suppose that the percentage of no-shows at flight time is 2%. For a particular flight with 197 seats, a total of 200 tickets were sold. What is the probability that the airline overbooked this flight?

Q6. The following is a portion of a classic data set called the “pilot plot data” in Fitting Equations to Data by Daniel and Wood, published in 1971. The response y is the acid content of material produced by titration, whereas the regressor x is the organic acid content produced by extraction and weighing.

y	x	y	x
76	123	70	109
62	55	37	48
66	100	82	138
58	75	88	164
88	159	43	28

Fit a simple linear regression; estimate a slope and intercept?

Card 8

Q1. Jointly distributed random variables. Examples

Q2. Hypothesis Testing. Examples

Q3. The density function of the continuous random variable X , the total number of hours, in units of 100 hours, that a family runs a vacuum cleaner over a period of one year, is given

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the average number of hours per year that families run their vacuum cleaners.

Q4. The number of tickets issued for traffic violations by 8 state troopers during the Memorial Day weekend are 5, 4, 7, 7, 6, 3, 8, and 6.

(a) If these values represent the number of tickets issued by a random sample of 8 state troopers from Montgomery County in Virginia, define a suitable population.

(b) If the values represent the number of tickets issued by a random sample of 8 state troopers from South Carolina, define a suitable population.

Q5. A random sample of 12 shearing pins is taken in a study of the Rockwell hardness of the pin head. Measurements on the Rockwell hardness are made for each of the 12, yielding an average value of 48.50 with a sample standard deviation of 1.5. Assuming the measurements to be normally distributed, construct a 90% confidence interval for the mean Rockwell hardness.

Q6. *John E. Croft* claims that over 40% of those who suffer from osteoarthritis receive measurable relief from an ingredient produced by a particular species of mussel found off the coast of New Zealand. To test this claim, the mussel extract is to be given to a group of 7 osteoarthritic patients. If 3 or more of the patients receive relief, we shall not reject the null hypothesis that $p = 0.4$; otherwise, we conclude that $p < 0.4$.

(a) Evaluate α , assuming that $p = 0.4$.

(b) Evaluate β for the alternative $p = 0.3$.

Q1. Definition of a random event, its notation. Examples

Q2. Testing means of a Normal Population. Examples

Q3. In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that

- (a) the student took mathematics or history;
- (b) the student did not take either of these subjects;
- (c) the student took history but not mathematics.

Q4. According to a survey one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is

- (a) anywhere from 2 to 5;
- (b) fewer than 3.

Q5. The loaves of rye bread distributed to local stores by a certain bakery have an average length of 30 centimeters and a standard deviation of 2 centimeters. Assuming that the lengths are normally distributed, what percentage of the loaves are

- (a) longer than 31.7 centimeters?
- (b) between 29.3 and 33.5 centimeters in length?
- (c) shorter than 25.5 centimeters?

Q6. A study was made by a retail merchant to determine the relation between weekly advertising expenditures and sales.

Advertising Costs (\$)	Sales (\$)
40	385
20	400
25	395
20	365
30	475
50	440
40	490
20	420
50	560
40	525
25	480
50	510

- (a) Find the equation of the regression line to predict weekly sales from advertising expenditures.
- (b) Estimate the weekly sales when advertising costs are \$35.

Q1. Theorem of the addition of probabilities

Q2. Maximum Likelihood Estimation. Examples

Q3. From a box containing 4 black balls and 2 green balls, 3 balls are drawn in succession, each ball being replaced in the box before the next draw is made. Find the probability distribution for the number of green balls.

Q4. In a chemical process, the amount of a certain type of impurity in the output is difficult to control and is thus a random variable. Speculation is that the population mean amount of the impurity is 0.20 gram per gram of output. It is known that the standard deviation is 0.1 gram per gram. An experiment is conducted to gain more insight regarding the speculation that $\mu = 0.2$. The process is run on a lab scale 50 times and the sample average $\bar{x}$ turns out to be 0.23 gram per gram. Comment on the speculation that the mean amount of impurity is 0.20 gram per gram.

Q5. Regular consumption of presweetened cereals contributes to tooth decay, heart disease, and other degenerative diseases. In a random sample consisting of 20 similar single servings of Alpha-Bits, the average sugar content was 11.3 grams with a standard deviation of 2.45 grams. Assuming that the sugar contents are normally distributed, construct a 95% confidence interval for the mean sugar content for single servings of Alpha-Bits.

Q6. A dry cleaning establishment claims that a new spot remover will remove more than 70% of the spots to which it is applied. To check this claim, the spot remover will be used on 12 spots chosen at random. If fewer than 11 of the spots are removed, we shall not reject the null hypothesis that $p = 0.7$; otherwise, we conclude that $p > 0.7$.

(a) Evaluate α , assuming that $p = 0.7$.

(b) Evaluate β for the alternative $p = 0.9$.

Q1. Classical definition of probability of an event. Examples

Q2. CI for Bernoulli random variable. Examples

Q3. Twenty adult males between the ages of 30 and 40 participated in a study to evaluate the effect of a specific health regimen involving diet and exercise on the blood cholesterol. Ten were randomly selected to be a control group, and ten others were assigned to take part in the regimen as the treatment group for a period of 6 months. The following data show the reduction in cholesterol experienced for the time period for the 20 subjects:

Control group:	7	3	-4	14	2
	5	22	-7	9	5
Treatment group:	-6	5	9	4	4
	12	37	5	3	3

Compute the sample variance and the sample standard deviation for both control and treatment groups?

Q4. Seventy new jobs are opening up at an automobile manufacturing plant, and 1000 applicants show up for the 70 positions. To select the best 70 from among the applicants, the company gives a test that covers mechanical skill, manual dexterity, and mathematical ability. The mean grade on this test turns out to be 60, and the scores have a standard deviation of 6. Can a person who scores 84 count on getting one of the Jobs?

Q5. A lawyer commutes daily from his suburban home to his midtown office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed.

- (a) What is the probability that a trip will take at least 1/2 hour?
- (b) If the office opens at 9:00 A.M. and the lawyer leaves his house at 8:45 A.M. daily, what percentage of the time is he late for work?
- (c) If he leaves the house at 8:35 A.M. and coffee is served at the office from 8:50 A.M. until 9:00 A.M., what is the probability that he misses coffee?
- (d) Find the length of time above which we find the slowest 15% of the trips.
- (e) Find the probability that 2 of the next 3 trips will take at least 1/2 hour.

Q6. The thrust of an engine (y) is a function of exhaust temperature (x) in °F when other important variables are held constant. Consider the following data.

y	x	y	x
4300	1760	4010	1665
4650	1652	3810	1550
3200	1485	4500	1700
3150	1390	3008	1270
4950	1820		

Fit a simple linear regression to the data

Q1. Conditional probability. Examples

Q2. Simple linear regression model. Examples

Q3. One prominent physician claims that 70% of those with lung cancer are chain smokers. If his assertion is correct,

(a) find the probability that of 10 such patients recently admitted to a hospital, fewer than half are chain smokers;

(b) find the probability that of 20 such patients recently admitted to a hospital, fewer than half are chain smokers.

Q4. A machine produces metal pieces that are cylindrical in shape. A sample of pieces is taken, and the diameters are found to be 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, and 1.03 centimeters. Find a 99% confidence interval for the mean diameter of pieces from this machine, assuming an approximately normal distribution.

Q5. A random sample of size 25 is taken from a normal population having a mean of 80 and a standard deviation of 5. A second random sample of size 36 is taken from a different normal population having a mean of 75 and a standard deviation of 3. Find the probability that the sample mean computed from the 25 measurements will exceed the sample mean computed from the 36 measurements by at least 3.4 but less than 5.9. Assume the difference of the means to be measured to the nearest tenth.

Q6. A random sample of 400 voters in a certain city are asked if they favor an additional 4% gasoline sales tax to provide badly needed revenues for street repairs. If more than 220 but fewer than 260 favor the sales tax, we shall conclude that 60% of the voters are for it.

(a) Find the probability of committing a type I error if 60% of the voters favor the increased tax.

(b) What is the probability of committing a type II error using this test procedure if actually only 48% of the voters are in favor of the additional gasoline tax?

Q1. Sample space. Examples

Q2. The t – Distribution. Examples

Q3. The probability that a vehicle entering the Luray Caverns has Canadian license plates is 0.12; the probability that it is a camper is 0.28; and the probability that it is a camper with Canadian license plates is 0.09. What is the probability that

- (a) a camper entering the Luray Caverns has Canadian license plates?
- (b) a vehicle with Canadian license plates entering the Luray Caverns is a camper?
- (c) a vehicle entering the Luray Caverns does not have Canadian plates or is not a camper?

Q4. *Fortune magazine (March 1997)* reported the total returns to investors for the 10 years prior to 1996 and also for 1996 for 431 companies. The total returns for 10 of the companies are listed below. Find a 95% confidence interval for the mean change in percent return to investors.

Company	Total Return to Investors	
	1986–96	1996
Coca-Cola	29.8%	43.3%
Mirage Resorts	27.9%	25.4%
Merck	22.1%	24.0%
Microsoft	44.5%	88.3%
Johnson & Johnson	22.2%	18.1%
Intel	43.8%	131.2%
Pfizer	21.7%	34.0%
Procter & Gamble	21.9%	32.1%
Berkshire Hathaway	28.3%	6.2%
S&P 500	11.8%	20.3%

Q5. The scores on a placement test given to college freshmen for the past five years are approximately normally distributed with a mean $\mu = 74$ and a variance $\sigma^2 = 8$. Would you still consider $\sigma^2 = 8$ to be a valid value of the variance if a random sample of 20 students who take the placement test this year obtain a value of $s^2 = 20$?

Q6. A new radar device is being considered for a certain missile defense system. The system is checked by experimenting with aircraft in which a kill or a no kill is simulated. If, in 300 trials, 250 kills occur, accept or reject, at the 0.04 level of significance, the claim that the probability of a kill with the new system does not exceed the 0.8 probability of the existing device.

Card 14

Q1. Bayes' rule.

Q2. Type I and Type II errors. Examples

Q3. In the senior year of a high school graduating class of 100 students, 42 studied mathematics, 68 studied psychology, 54 studied history, 22 studied both mathematics and history, 25 studied both mathematics and psychology, 7 studied history but neither mathematics nor psychology, 10 studied all three subjects, and 8 did not take any of the three. Randomly select a student from the class and find the probabilities of the following events.

(a) A person enrolled in psychology takes all three subjects.

(b) A person not taking psychology is taking both history and mathematics.

Q4. A private pilot wishes to insure his airplane for \$200,000. The insurance company estimates that a total loss will occur with probability 0.002, a 50% loss with probability 0.01, and a 25% loss with probability 0.1. Ignoring all other partial losses, what premium should the insurance company charge each year to realize an average profit of \$500?

Q5. The distribution of heights of a certain breed of terrier has a mean of 72 centimeters and a standard deviation of 10 centimeters, whereas the distribution of heights of a certain breed of poodle has a mean of 28 centimeters with a standard deviation of 5 centimeters. Assuming that the sample means can be measured to any degree of accuracy, find the probability that the sample mean for a random sample of heights of 64 terriers exceeds the sample mean for a random sample of heights of 100 poodles by at most 44.2 centimeters.

Q6. A manufacturer has developed a new fishing line, which the company claims has a mean breaking strength of 15 kilograms with a standard deviation of 0.5 kilogram. To test the hypothesis that $\mu = 15$ kilograms against the alternative that $\mu < 15$ kilograms, a random sample of 50 lines will be tested. The critical region is defined to be $\bar{x} < 14.9$.

(a) Find the probability of committing a type I error when H_0 is true.

(b) Evaluate β for the alternatives $\mu = 14.8$ and $\mu = 14.9$ kilograms.

Card 15

Q1. Definition of multiplication of events. Examples

Q2. The Chi-Squared Distribution. Examples

Q3. For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15. What is the probability that

- (a) at least one member of a married couple will vote?
- (b) a wife will vote, given that her husband will vote?
- (c) a husband will vote, given that his wife will not vote?

Q4. If the probability that a fluorescent light has a useful life of at least 800 hours is 0.9, find the probabilities

that among 20 such lights

- (a) exactly 18 will have a useful life of at least 800 hours;
- (b) at least 15 will have a useful life of at least 800 hours;
- (c) at least 2 will not have a useful life of at least 800hours.

Q5. In a biomedical research study, it was determined that the survival time, in weeks, of an animal subjected to a certain exposure of gamma radiation has a gamma distribution with $\alpha = 5$ and $\beta = 10$.

- (a) What is the mean survival time of a randomly selected animal of the type used in the experiment?
- (b) What is the standard deviation of survival time?
- (c) What is the probability that an animal survives more than 30 weeks?

Q6. A study of the amount of rainfall and the quantity of air pollution removed produced the following

Daily Rainfall, x (0.01 cm)	Particulate Removed, y ($\mu\text{g}/\text{m}^3$)
4.3	126
4.5	121
5.9	116
5.6	118
6.1	114
5.2	118
3.8	132
2.1	141
7.5	108

- (a) Find the equation of the regression line to predict the particulate removed from the amount of daily rainfall.
- (b) Estimate the amount of particulate removed when the daily rainfall is $x = 4.8$ units.

Q1. Dependent and independent events. Examples

Q2. The Central Limit Theorem. Examples

Q3. An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

Find

- (a) $P(T = 5)$;
- (b) $P(T > 3)$;
- (c) $P(1.4 < T < 6)$;
- (d) $P(T \leq 5 \mid T \geq 2)$.

Q4. The mean score for freshmen on an aptitude test at a certain college is 540, with a standard deviation of 50. Assume the means to be measured to any degree of accuracy. What is the probability that two groups selected at random, consisting of 32 and 50 students, respectively, will differ in their mean scores by

- (a) more than 20 points?
- (b) an amount between 5 and 10 points?

Q5. The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

3.4	2.5	4.8	2.9	3.6
2.8	3.3	5.6	3.7	2.8
4.4	4.0	5.2	3.0	4.8

Assuming that the measurements represent a random sample from a normal population, find a 95% prediction interval for the drying time for the next trial of the paint.

Q6. A soft-drink machine at a steak house is regulated so that the amount of drink dispensed is approximately normally distributed with a mean of 200 milliliters and a standard deviation of 15 milliliters. The machine is checked periodically by taking a sample of 9 drinks and computing the average content. If $\bar{x}$ falls in the interval $191 < \bar{x} < 209$, the machine is thought to be operating satisfactorily; otherwise, we

conclude that $\mu \neq 200$ milliliters.

- (a) Find the probability of committing a type I error when $\mu = 200$ milliliters.
- (b) Find the probability of committing a type II error when $\mu = 215$ milliliters.

Card 17

Q1. Normal Random Variable. Examples

Q2. Significance levels. Examples

Q3. An attendant at a car wash is paid according to the number of cars that pass through. Suppose the probabilities are $1/12$, $1/12$, $1/4$, $1/4$, $1/6$, and $1/6$, respectively, that the attendant receives \$7, \$9, \$11, \$13, \$15, or \$17 between 4:00 P.M. and 5:00 P.M. on any sunny Friday. Find the attendant's expected earnings for this particular period.

Q4. 4 million workers in the general workforce, 5.8% tested positive for drugs. Of those testing positive, 22.5% were cocaine users and 54.4% marijuana users.

(a) What is the probability that of 10 workers testing positive, 2 are cocaine users, 5 are marijuana users, and 3 are users of other drugs?

(b) What is the probability that of 10 workers testing positive, all are marijuana users?

(c) What is the probability that of 10 workers testing positive, none is a cocaine user?

Q5. A random sample of 12 graduates of a certain secretarial school typed an average of 79.3 words per minute with a standard deviation of 7.8 words per minute. Compute the 95% prediction interval for the next observed number of words per minute typed by a graduate of the secretarial school.

Q6. A study was made by a retail merchant to determine the relation between weekly advertising expenditures and sales.

Advertising Costs (\$)	Sales (\$)
40	385
20	400
25	395
20	365
30	475
50	440
40	490
20	420
50	560
40	525
25	480
50	510

Test the hypothesis that $\beta_1 = 6$ against the alternative that $\beta_1 < 6$. Use a 0.025 level of significance.

Card 18

Q1. Definition of the sum of events. Examples

Q2. The Uniform Random Variable. Examples

Q3. A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

(a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?

(b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

Q4. Suppose that the probabilities are 0.4, 0.3, 0.2, and 0.1, respectively, that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year. Find the mean and variance of the random variable X representing the number of power failures striking this subdivision.

Q5. A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation $\sigma_1 = 5$, has a mean $\bar{x}_1 = 80$. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation $\sigma_2 = 3$, has a mean $\bar{x}_2 = 75$. Find a 94% confidence interval for $\mu_1 - \mu_2$.

Q6. A vote is to be taken among the residents of a town and the surrounding county to determine whether a proposed chemical plant should be constructed. The construction site is within the town limits, and for this reason many voters in the county believe that the proposal will pass because of the large proportion of town voters who favor the construction. To determine if there is a significant difference in the proportions of town voters and county voters favoring the proposal, a poll is taken. If 120 of 200 town voters favor the proposal and 240 of 500 county residents favor it, would you agree that the proportion of town voters favoring the proposal is higher than the proportion of county voters? Use an $\alpha = 0.05$ level of significance.

Card 19

Q1. Definition of probability of an event, its notation. Examples

Q2. Least squares and the fitted model. Examples

Q3. A coin is tossed 400 times. Use the normal curve approximation to find the probability of obtaining

- (a) between 185 and 210 heads inclusive;
- (b) exactly 205 heads;
- (c) fewer than 176 or more than 227 heads.

Q4. An insurance company offers its policyholders a number of different premium payment options. For a randomly selected policyholder, let X be the number of months between successive payments. The cumulative distribution function of X is

$$F(x) = \begin{cases} 0, & \text{if } x < 1, \\ 0.4, & \text{if } 1 \leq x < 3, \\ 0.6, & \text{if } 3 \leq x < 5, \\ 0.8, & \text{if } 5 \leq x < 7, \\ 1.0, & \text{if } x \geq 7. \end{cases}$$

- (a) What is the probability mass function of X ?
- (b) Compute $P(4 < X \leq 7)$.

Q5. A type of thread is being studied for its tensile strength properties. Fifty pieces were tested under similar conditions, and the results showed an average tensile strength of 78.3 kilograms and a standard deviation of 5.6 kilograms. Assuming a normal distribution of tensile strengths, give a lower 95% prediction limit on a single observed tensile strength value. In addition, give a lower 95% tolerance limit that is exceeded by 99% of the tensile strength values.

Q6. The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows:

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

- (a) Estimate the linear regression line.
- (b) Estimate the final examination grade of a student who received a grade of 85 on the midterm report.

Card 20

Q1. Formula of a total probability.

Q2. Confidence intervals for a normal mean. Examples

Q3. A manufacturing company uses an acceptance scheme on items from a production line before they are shipped. The plan is a two-stage one. Boxes of 25 items are readied for shipment, and a sample of 3 items is tested for defectives. If any defectives are found, the entire box is sent back for 100% screening. If no defectives are found, the box is shipped.

- (a) What is the probability that a box containing 3 defectives will be shipped?
- (b) What is the probability that a box containing only 1 defective will be sent back for screening?

Q4. If the number of hurricanes that hit a certain area of the eastern United States per year is a random variable having a Poisson distribution with $\mu = 6$, find the probability that this area will be hit by

- (a) exactly 15 hurricanes in 2 years;
- (b) at most 9 hurricanes in 2 years.

Q5. Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

Q6. A study was conducted at Virginia Tech to determine if certain static arm-strength measures have an influence on the “dynamic lift” characteristics of an individual. Twenty-five individuals were subjected to strength tests and then were asked to perform a weightlifting test in which weight was dynamically lifted overhead. The data are given here.

Individual	Arm Strength, x	Dynamic Lift, y
1	17.3	71.7
2	19.3	48.3
3	19.5	88.3
4	19.7	75.0
5	22.9	91.7
6	23.1	100.0
7	26.4	73.3
8	26.8	65.0
9	27.6	75.0
10	28.1	88.3
11	28.2	68.3
12	28.7	96.7
13	29.0	76.7
14	29.6	78.3
15	29.9	60.0
16	29.9	71.7
17	30.3	85.0
18	31.3	85.0
19	36.0	88.3
20	39.5	100.0
21	40.4	100.0
22	44.3	100.0
23	44.6	91.7
24	50.4	100.0
25	55.9	71.7

- (a) evaluate s^2 ;
- (b) test the hypothesis that $\beta_1 = 0$ against the alternative that $\beta_1 \neq 0$ at the 0.05 level of significance and interpret the resulting decision.

Card 21

Q1. Continuous random variables. Examples

Q2. The F - Distribution. Examples

Q3. Biologists doing studies in a particular environment often tag and release subjects in order to estimate the size of a population or the prevalence of certain features in the population. Ten animals of a certain population thought to be extinct (or near extinction) are caught, tagged, and released in a certain region. After a period of time, a random sample of 15 of this type of animal is selected in the region. What is the probability that 5 of those selected are tagged if there are 25 animals of this type in the region?

Q4. Suppose the random variable X follows a beta distribution with $\alpha = 1$ and $\beta = 3$.

(a) Determine the mean and median of X .

(b) Determine the variance of X .

(c) Find the probability that $X > 1/3$.

Q5. A manufacturer of car batteries guarantees that the batteries will last, on average, 3 years with a standard deviation of 1 year. If five of these batteries have lifetimes of 1.9, 2.4, 3.0, 3.5, and 4.2 years, should the manufacturer still be convinced that the batteries have a standard deviation of 1 year? Assume that the battery lifetime follows a normal distribution.

Q6. The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows:

x	77	50	71	72	81	94	96	99	67
y	82	66	78	34	47	85	99	99	68

(a) evaluate s^2 ;

(b) construct a 95% confidence interval for β_0 ;

(c) construct a 95% confidence interval for β_1 .

Card 22

Q1. CI for Bernoulli random variable. Examples

Q2. The Chi-Squared Distribution. Examples

Q3. Suppose that a grocery store purchases 5 cartons of skim milk at the wholesale price of \$1.20 per carton and retails the milk at \$1.65 per carton. After the expiration date, the unsold milk is removed from the shelf and the grocer receives a credit from the distributor equal to three-fourths of the wholesale price. If the probability distribution of the random variable X , the number of cartons that are sold from this lot, is

x	0	1	2	3	4	5
$f(x)$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{4}{15}$	$\frac{3}{15}$

find the expected profit.

Q4. A telemarketing company has a special letter opening machine that opens and removes the contents of an envelope. If the envelope is fed improperly into the machine, the contents of the envelope may not be removed or may be damaged. In this case, the machine is said to have “failed.”

- If the machine has a probability of failure of 0.01, what is the probability of more than 1 failure occurring in a batch of 20 envelopes?
- If the probability of failure of the machine is 0.01 and a batch of 500 envelopes is to be opened, what is the probability that more than 8 failures will occur?

Q5. An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.

Q6. A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took the regular course were recorded.

- Find the equation of the regression line to predict course grades from placement test scores.
- If 60 is the minimum passing grade, below which placement test score should students in the future

be denied admission to this course?

Placement Test	Course Grade
50	53
35	41
35	61
40	56
55	68
65	36
35	11
60	70
90	79
35	59
90	54
80	91
60	48
60	71
60	71
40	47
55	53
50	68
65	57
50	79

Card 23

Q1. Discrete random variables. Examples

Q2. Testing means of a Normal Population. Examples

Q3. Pollution of the rivers in the United States has been a problem for many years. Consider the following events:

A: the river is polluted,

B: a sample of water tested detects pollution,

C : fishing is permitted

Assume $P(A) = 0.3$, $P(B|A) = 0.75$, $P(B|A') = 0.20$,
 $P(C|A \cap B) = 0.20$, $P(C|A' \cap B) = 0.15$, $P(C|A \cap B') = 0.80$, and $P(C|A' \cap B') = 0.90$.

(a) Find $P(A \cap B \cap C)$.

(b) Find $P(B' \cap C)$.

(c) Find $P(C)$.

(d) Find the probability that the river is polluted, given that fishing is permitted and the sample tested did not detect pollution.

Q4. A random sample of 5 bank presidents indicated annual salaries of \$395,000, \$521,000, \$483,000, \$479,000, and \$510,000. Find the variance of this set.

Q5. A study was conducted by the Department of Zoology at the Virginia Tech to estimate the difference in the amounts of the chemical ortho phosphorus measured at two different stations on the James River. Ortho phosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average ortho phosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligram per liter. Find a 95% confidence interval for the difference in the true average ortho phosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

Q6. A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that $\mu = 8$ kilograms against the alternative that $\mu \neq 8$ kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

Card 24

Q1. Properties of probability.

Q2. Inferences for regression coefficients. Examples

Q3. In a certain region of the country it is known from past experience that the probability of selecting an adult over 40 years of age with cancer is 0.05. If the probability of a doctor correctly diagnosing a person with cancer as having the disease is 0.78 and the probability of incorrectly diagnosing a person without

cancer as having the disease is 0.06, what is the probability that an adult over 40 years of age is diagnosed as having cancer?

Q4. Suppose that the error in the reaction temperature, in $^{\circ}\text{C}$, for a controlled laboratory experiment is a continuous random variable X having the probability density function

(a) Verify that $f(x)$ is a density function.

(b) Find $P(0 < X \leq 1)$.

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Q5. If the proportion of a brand of television set requiring service during the first year of operation is a random variable having a beta distribution with $\alpha = 3$ and $\beta = 2$, what is the probability that at least 80% of the new models of this brand sold this year will require service during their first year of operation?

Q6. The following data were collected to determine the relationship between pressure and the corresponding scale reading for the purpose of calibration.

Pressure, x (lb/sq in.)	Scale Reading, y
10	13
10	18
10	16
10	15
10	20
50	86
50	90
50	88
50	88
50	92

(a) Find the equation of the regression line.

(b) The purpose of calibration in this application is to estimate pressure from an observed scale reading. Estimate the pressure for a scale reading of 54 using $\hat{x} = (54 - b_0)/b_1$.

Card 25

Q1. Bayes' rule.

Q2. Simple linear regression model. Examples

Q3. Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.

Q4. The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X;
- (b) using the probability density function of X.

Q5. Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute.

Q6. In a certain type of metal test specimen, the normal stress on a specimen is known to be functionally related to the shear resistance. The following is a set of coded experimental data on the two variables:

Normal Stress, x	Shear Resistance, y
26.8	26.5
25.4	27.3
28.9	24.2
23.6	27.1
27.7	23.6
23.9	25.9
24.7	26.3
28.1	22.5
26.9	21.7
27.4	21.4
22.6	25.8
25.6	24.9

- (a) Estimate the regression line $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (b) Estimate the shear resistance for a normal stress of 24.5.